Analyzing and Predicting Stock Market Trends using Historical Market Data

(COMP3125 Individual Project)

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***Abstract*—This project analyzes five years of historical data for S&P 500 companies (2013-2018) to analyze and predict market behavior, focusing on identifying trends, evaluating performance, and modeling future price movements. It investigates growth and variability across stocks and evaluates the use of machine learning models to predict next day closing prices. Technical indicators such as moving averages, RSI, and momentum are used to enhance model performance. Results reveal that predictive models can provide valuable insight for informed investment decisions.**

Keywords—example1, example2, example3, example 4, example 5 (provide 3-5 keywords)

I. Introduction (*Heading 1*)

Analyzing and predicting stock price movement has been a widely studied problem in both the finance and machine learning sectors. Creating models with accurate short-term predictions can support trading strategies, risk and portfolio management. Using the historical data for S&P 500 companies (2013-2018) the project (1) analyzes the companies with the most growth and regression, (2) examines the relationship between stock volatility/trading volume and consistent upward trends, (3) analyze which stocks have returned most consistent to the market average (10%), and (4) develop models to predict the next day closing price using the data set and technical indicators.

II. Datasets

A. Kaggle (S&P 500 Data 2013-2018)

The dataset used in our analysis and used to train the models is from Kaggle, S&P 500 Data 2013-2018 which is a dataset containing daily OHLCV (Open, High, Low, Close, Volume) for S&P 500 constituents spanning January 1, 2013 – December 31, 2018. The dataset was downloaded from Kaggle, a reputable website known for its reliable datasets for machine learning and data science. The dataset was cleaned to remove erroneous entries and non-trading days.

B. Character of the datasets

The dataset is stored in a CSV format with approximately 1,000,000 rows and the following columns:

TABLE 1: DATASET CHARACTERISTICS

Dataset Characteristics		
Column	**Description**	**Unit**
Date	Trading Date	YYYY-MM-DD
Open	Price at market open	USD
High	Highest price of day	USD
Low	Lowest price of day	USD
Close	Price at market close	USD
Volume	Number of shares traded	Shared
Name	Stock ticker	---

The dataset was cleaned by removing erroneous entries and non-trading days. The technical indicators were calculated using the provided data then added to the data set.

C. Technical Indicators

Technical Indicators were calculated using the dataset and then added back to the data set to provide more robust data for training the models. The technical indicators include.

- Moving Averages (7-day, 30-day, 90-day)

$$MA_n = \frac{\sum_{i=0}^{n} closing_n}{n}$$

- Volatility (30-day)

$$\sigma_{30}(t) = \sqrt{[1/(n-1) \times \Sigma_{i=0}^{n-1} (P_{t\,i}^{-} - \mu_{30})^2]}$$

$$\mu_{30} = (1/30) \times \Sigma_{i=0}^{29} P_{t-i}$$

P_t = Closing price at time t

- Momentum (10-day)

$$M_{10}(t) = P_t - P_{t-10}$$

P_t = Closing price at time t

- Volume Average (30-day)

$$V_{30}(t) = (1/30) \times \Sigma_{i=0}^{29} V_{t-i}$$

V_{t-i} = trading volume at time (t-i)

- Relative Strength Index (RSI) (14-day)

$$RSI = 100 - \left(\frac{100}{(1+RS)}\right)$$

$$RS = \frac{Avg\ Gain}{Avg\ Loss}$$

III. METHODOLOGY

In this part, you should give an introduction of the methods/model. First, what's the method/model. What's the assumption of this method/model. What's the advantage/disadvantage of this method/model. Why did you choose it. What Python module or function do you apply to apply this method/model. Any optional input/extra work did you adjust to make the results better. If you have multiple methods, feel free to use subsection A., B. to separate them.

The dataset is imported using the pandas module using the read_csv function, and dates are parsed into datetime format for time series operations. The dataset is sorted by company name (Name) and date to ensure chronological order. Technical indicators are then calculated and added into new columns in the dataset.

A. Company Growth and Volatility Analysis

This analysis focused on comprehensive company growth and volatility profiling. For each company, growth over the full dataset is calculated as:

$$Growth = \frac{Close_{end} - Close_{start}}{Close_{start}} * 100\%$$

Daily volatility is measured as the standard deviation of daily returns, which is measuring the dispersion of returns which serves as a risk estimate. The more volatile the asset the riskier the asset is. Once the calculations are finished, the dataset is sorted once again, this time by growth, revealing the top 5 companies with the most growth and the bottom 5 companies with the worst growth.

B. Stock Volatility and Trading Volume Analysis

This analysis of the dataset focused on rolling window volatility and trend analysis to capture short term market dynamics. To balance noise reduction and market responsiveness, a 30-day window was chosen, within four metrics were calculated. (1) Volatility, (2) average volume, (3) 30-day return, and (4) trend indicators showing net price movement direction. Both volatility and volume were categorized into quartiles, transforming the measures into ordinal categories to enable analysis without linear relationships. A trend probability was constructed, calculating mean and standard deviation of returns per volatility-volume category alongside empirical probability of upward trends.

C. Annual Return Analysis

This analysis concentrates on the annualized return computation and comprehensive stability analysis through grouping and ranking. After importing the dataset and organizing chronologically, company specific metric was calculated using grouped aggressions, including first and last closing prices and closing price variance. Total returns were computed as the percentage change between initial and final closing prices, then converted into annualized form using the compound growth formula.

$$(1 + total\ return)^{\frac{1}{4}} - 1$$

Identify applicable funding agency here. If none, delete this text box.

Assuming a four year investment period to enable comparability with common investment benchmarks, mainly the S&P 500 long term average of approximately 10%. Target return was assessed by computing the absolute difference between each stocks annual return and the 10% benchmark, creating a quantitative measure for ranking stocks. Two different analytical perspectives were employed for stability assessment, first identifying the top 10 stocks closest to the return, and second filtering stocks within a 7.5% to 12.5% annual return range and ranking the top ten by lowest variance to identify the most stable returners.

D. Predictive Model Development for Next Day Closing Price

The predictive model developed using feature engineering and algorithmic comparisons to predict the next day closing price. After sorting chronologically using company name and date, technical indicators computation was performed on a per stock basis, through iterative processing. The expanded feature set included seven technical indicators: 7-day, 30-day, and 90-day moving averages, 30-day rolling volatility, 10-day momentum, 30-day average volume, and 14-day RSI (Relative Strength Index). Basic features included open, high, low, volume, and previous day's closing price, while the combined feature set merged both the basic and technical indicators for a comparative analysis. Individual stock processing filtered the dataset removing stocks with less than 500 observations and then splitting the dataset into a 60%-40% train-test split. Five different algorithms were evaluated:
Linear Regression serves as a baseline linear model as is implemented through sklearn library. The Ridge Regression with L2 regularization addresses multicollinearity by shrinking coefficients towards zero, which is powerful when technical indicators exhibit correlation. Lasso Regression with L1 regularization performs automatic feature selection, identifying the most predictive indicators but struggles when multiple features are equally predictive. Random Forest encapsulates non-linear relationships through ensembles decision trees, providing performance against outliers, but is prone to overfitting with a limited training set. Gradient Boosting employs sequential tree building to correct previous errors, achieving performance through adaptive learning, but is prone to overfitting if not tuned correctly and is computationally expensive. All the models were trained though the sklearn library. Separate StandardScaler objects were applied to both the basic and full feature sets to normalize inputs to zero mean and unit variance for scale sensitive model, ensuring each feature contributes equally to model learning.

IV. RESULTS

A. Result A

Growth analysis across the dataset demonstrated a diverse spectrum, with total returns ranging from 1749% to -86%. The results showed that higher volatility predicts lower returns as seen on the scatterplot and boxplot below.

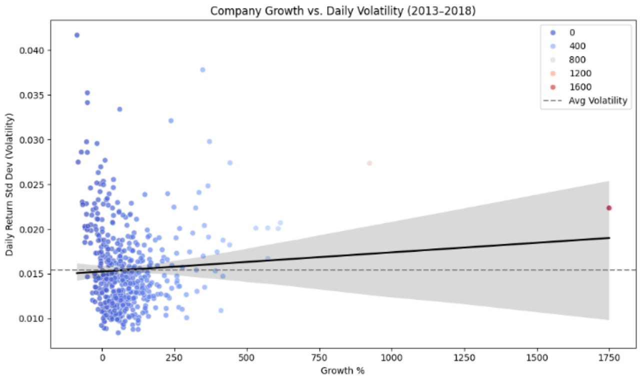

Figure 1: Company Growth Versus Daily Volatility

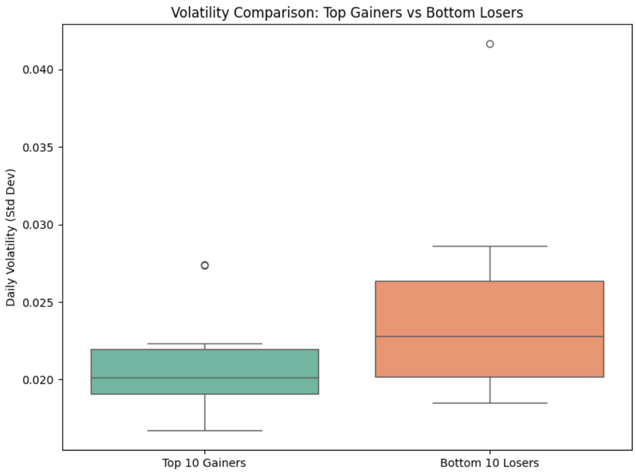

Figure 2: Volatility Comparison Top Gainers Versus Top Losers

Below is the analysis of best and worst performers in the dataset.

Top 5 Growing Companies		
Name	**Growth (%)**	**Volatility**
NVDA	1749.64	0.0224
NFLX	923.33	0.0273
ALGN	615.95	0.0207
EA	608.41	0.0201
STZ	572.37	0.0166

Top 5 Regressing Companies		
Name	**Growth (%)**	**Volatility**
DISCK	-66.02	0.0227
DISCA	-67.65	0.0230
UA	-70.77	0.0286
RRC	-81.81	0.0275
CHK	-85.71	0.0417

B. Results B

Rolling window analysis across 30-day periods captured dynamic market behaviors characterized by different volatility-volume combinations. The volatility categorization into quartiles showed performance differences, with the Mid-Low Volatility quartile showing the highest upward trend at 60.57%, and the Low Volatility Quartile with the lowest upward trend at 59.15%. The volume categorization into quartiles also showed performance differences with the Low Volume quartile having the highest upward trend with 63.87% and the High-Volume quartile with the lowest upward trend with 56.72%. Boxplot visualizations below demonstrate return distribution between volatility and volume.

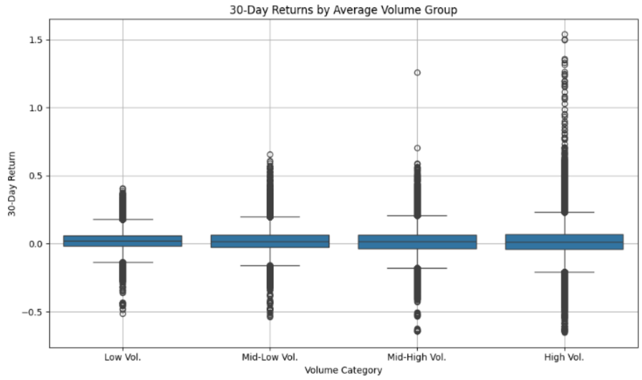

Figure 3: 30-Day Returns by Average Volume Group

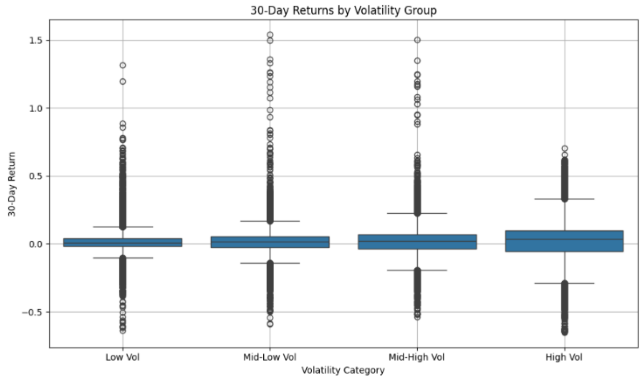

Figure 4: 30-Day Returns by Average Volatility Group

Trends analysis combining both the volatility quartiles and volume quartile provided great insight into upward trends. Concluding that the most favorable market conditions occurred during periods of low volume and mid-high volatility, while the worst market conditions occurred during periods of high volume and high volatility. All quartile combinations and their results can be seen in the heatmap below.

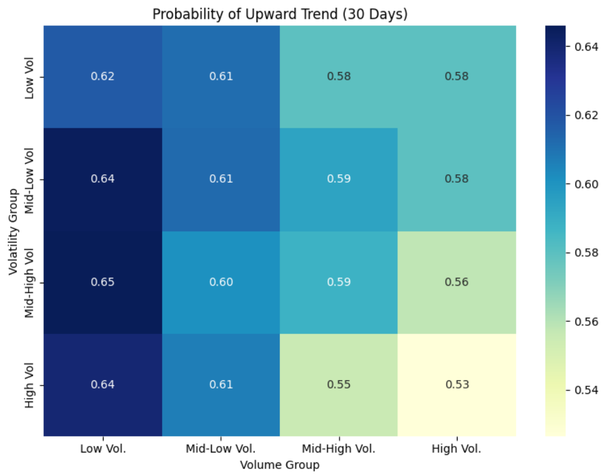

Figure 5: Probability of Upward Trend (30 Days)

heads.

C. Results C

Annualized return calculations to identify stocks with returns closest to the market average of 10%, while also considering price stability analyzing the distribution of stocks comparing their annual returns to the volatility, and found the top 10 most stable stocks with mid-range performance. The bar plot below highlights the most stable stocks with mid-range performance (7.5%-12.5%) annual return.

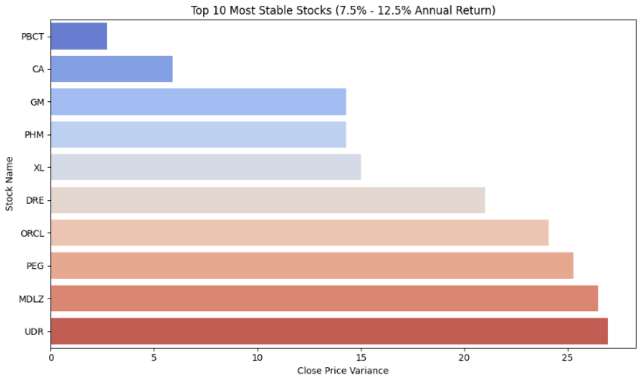

Figure 6: Top 10 Most Stable Stocks

When comparing all annual returns versus variance, results indicated a concentration between 5% and 15% returns with a noticeable spread beyond those bound. The distribution graph has high similarities to the bell curve, indicating normal distribution, suggesting the majority of the S&P stocks over the 5-year period delivered the approximated return (10%) while more outliers outperforming than underperforming.

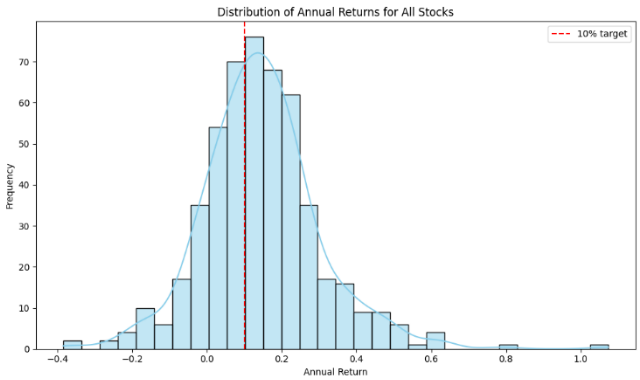

Figure 7: Distribution of Annual Returns of All Stocks

Model performance comparison across models revealed major differences in predictive models. The best performing models were the Ridge V3 (trained using alpha equal to 0.01), linear V1 (trained using 1st degree polynomial), and Ridge V2 (trained using alpha equal to 0.1). Below is a table with the top 5 models and an analysis of each. The strongest models were clearly the lasso and ridge model, which had significantly lower RMSE values than the other models, which represents the absolute average error the model had when predicting the next day closing price. Comparing median RMSE values with the average RMSE values we also see that outliers predictions greatly increasing the models average, therefore the median RMSE values are a better predictor than the average.

TABLE 2 TOP 5 MODELS

Top 5 Models			
Model Version	Avg_RMSE	Std RMSE	Median RMSE
Ridge V3	1.449077	1.447872	1.088306
Linear V1	1.449110	1.447495	1.088317
Ridge V2	1.450083	1.452446	1.089007
Lasso V3	1.461058	1.444442	1.092390
Lasso V2	1.470800	1.447581	1.103813

Visualization results provide more insight into the Ridge V3 model and its predictive accuracy. The boxplot and distribution below visualize how most of the models' predictions lie near zero, indicating an accurate result. Analyzing the Ridge V3 model more, the most important predictor for this model is high with an importance of 1.2727 followed by the low with an importance of 1.2362 and open with an importance of 0.7758. Below is a chart depicting the top fifteen predictors for the Ridge V3 model with their importance.

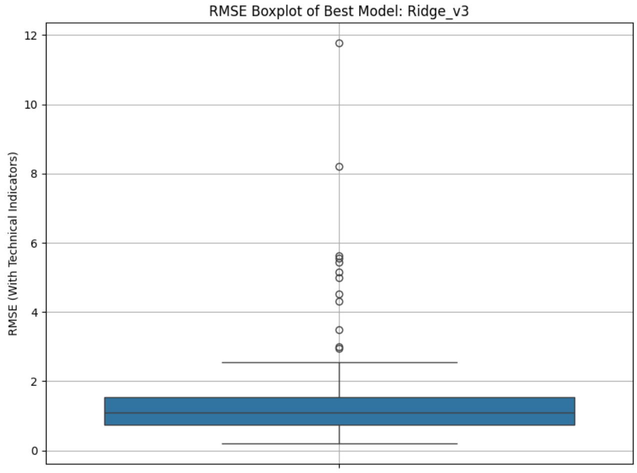

Figure 8: RMSE Boxplot of Ridge V3 Model

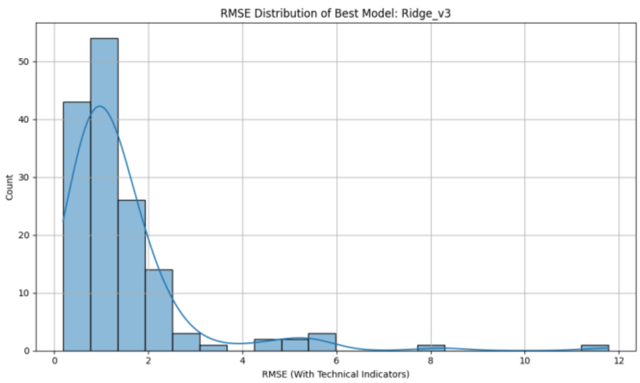

Figure 9: RMSE Distribution of Ridge V3 Model

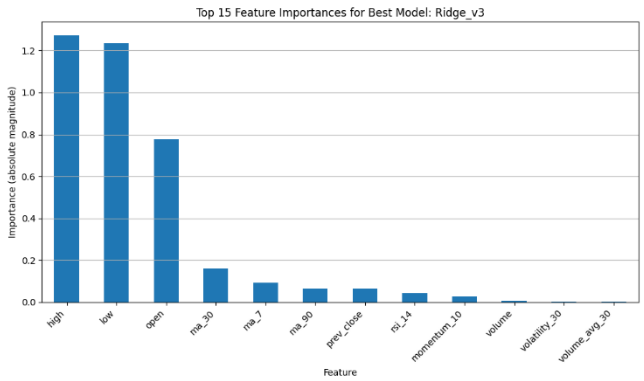

Figure 10: Feature Importance for Ridge V3 Model

V. Discussion

In my project there are several methodological limitations and areas for improvement, including expanding the dataset to include more stock and cover a larger period. With the limited dataset the models and analysis produced are limited generalized findings of that time and are biased towed the economic period. Another area of improvement is with the technical indicators and lack thereof, while the technical indicators were grounded based on financial theory, they represent only a subset of all relevant market dynamics. Future research directions should address these limitations through expanding the technical indicators used and the dataset to include a larger time period, more financial data and geopolitical data as it has a large impact of financial markets.

VI. Conclusion

In conclusion the analysis and development of predictive models over a five-year period has yielded important findings for quantitative finance. The evaluation of technical indicators to generate a predictive model is evidence that feature engineering can meaningfully enhance stock price prediction accuracy. The risk-return profiling identified optimal investing opportunities though the quantification of the relationship between volatility and performance. The rolling window analysis provided insights into market dynamics, demonstrating that combinations of volatility and volume create predictable conditions for positive price movements. These findings contribute to a broader understanding of the market and stand as evidence that with a systematic application of technical indicators, improvements and analysis can be made into complex market relationships that can inform both academic research and practical investment applications.